

—GENERAL SETUP—

I. Configuring Raspberry Pi's in Base Format

(Pi's 1, 2, 3, 4, 5 | *Raspberry Pi OS Lite* "2025-10-01-raspios-trixie-arm64-lite.img.xz")

For this setup, I am using a Mac running Sonoma 14.1 and Raspberry Pi Imager v1.8.5

Section 1: Configure the OS file on imaging device

1. Flash the storage device with image
"2025-10-01-raspios-trixie-arm64-lite.img.xz"



- Raspberry Pi Device
 - In this simulation I used Raspberry Pi 4's
 - Operating System
 - Choose the most recent Pi OS on the website
 - Select "Use Custom"
 - Storage
 - Choose SD Card inserted into device
2. Remove SD Card and Reinsert
 3. Open Terminal (commands for mac)

```
ls /volumes
cd /volumes/bootfs
touch ssh
```

 - This is to create and SSH file to enable SSH abilities

Section 2.1: Configuring Raspberry Pi on physical unit for IP/SSH

1. Insert SD Card into Pi and power up

- a. Connect Micro HDMI and Keyboard

2. Set username/password

USER: admin

PW: admin

3. Log into Pi on commandline

4. Enable SSH

```
sudo systemctl enable ssh
```

```
sudo systemctl start ssh
```

```
sudo systemctl status ssh
```

If done properly you should see a green "active (running)" after final command

5. Confirm that your Pi OS is using "Network Manager"

- a. Older Pi's used something called "dhcpcd", the root still exists but its not an executable service.

```
systemctl list-unit-files
```

6. Set IP Address

****Caps matter****

```
nmcli connection show
```

```
sudo nmcli connection modify "Wired connection 1" ipv4.addresses IP  
ADDR/Subnet Prefix
```

```
sudo nmcli connection modify "Wired connection 1" ipv4.gateway Router IP Addr
```

```
sudo nmcli connection modify "Wired connection 1" ipv4.dns "8.8.8.8"
```

```
sudo nmcli connection modify "Wired connection 1" ipv4.method manual
```

```
sudo nmcli connection up "Wired connection 1"
```

7. Confirm

```
Ifconfig
```

- a. Ping from computer to ensure you have a successful ip address change/

8. Test SSH

- a. On the terminal on your computer SSH
 - i. Ensure you are ssh'ing @ admin
>ssh admin@x.x.x.x

(Optional) Section 2.2: Configuring Raspberry Pi on physical unit for Wifi

This is optional because VM Router will route NIC's to internet after further setup, causes some issues down the road

1. Unlock RFKill

a. Open Raspi-Config

****No need to input your Wifi SSID or Password****

`Sudo Raspi-Config`

System Options>Wireless LAN>Enter Country Code

b. Double check RFKill is off

`rfkill list`

Should read:

0: hci0: Bluetooth

Soft blocked: yes

Hard blocked: no

1: phy0: Wireless LAN

Soft blocked: no

Hard blocked: no

2. Enable Network Manager

`systemctl status NetworkManager`

If inactive:

`sudo systemctl enable --now NetworkManager`

3. Enable Wifi & Scan

`sudo nmcli radio wifi on`

`sudo ip link set wlan0 up`

`sudo nmcli device wifi list`

##Alternative## `sudo iwlist wlan0 scan | grep ESSID**`

4. Confirm both interfaces are on

`ip addr show | egrep 'eth0|wlan0'`

5. Connect Wifi

`sudo nmcli device wifi connect "Network name" password "password"`

If error: 802-11-wireless-security.key-mgmt: property is missing

`sudo nmcli connection add type wifi ifname wlan0 con-name
"Network Name" ssid "Network Name"`

`sudo nmcli connection modify "Network Name"
802-11-wireless-security.key-mgmt wpa-psk`

`sudo nmcli connection modify "Network Name"
802-11-wireless-security.psk "password"`

`sudo nmcli connection up "Network Name"`

6. Check IP and Gateway

- a. you should see your network
nmcli device show wlan0

****At this point you'll be connected to Wifi but you still need to fix internal internet routing****

7. Fixing Dual-NIC Routing

This prevents eth0 from being the default internet route and assigns wlan0 as path

```
sudo nmcli connection modify "Wired connection 1"  
ipv4.never-default yes  
sudo nmcli connection modify "Wired connection 1"  
ipv6.never-default yes  
sudo nmcli connection modify "yournetworkname"  
ipv4.route-metric 50  
sudo nmcli connection reload  
sudo nmcli device reapply wlan0
```

8. Test

Ping gateway
Ping www.google.com

****Often times you can ping your gateway but not google, if this is the case, override DNS in step 9****

9. DNS and Internet Validation

```
sudo nmcli connection modify "Your Network"  
ipv4.ignore-auto-dns yes  
sudo nmcli connection modify "Your Network"  
ipv6.ignore-auto-dns yes  
sudo nmcli connection modify "Your Network" ipv4.dns  
"8.8.8.8 1.1.1.1"  
sudo nmcli connection up "Your Network"
```

10. Make these changes permanent

```
Sudo nmcli connection modify "Wired connection 1"  
connection.autoconnect yes  
sudo nmcli connection modify "Your Network"  
connection.autoconnect yes  
Sudo Reboot
```

11. (troubleshooting) Deleting UUID

```
nmcli connectuion show | grep "Your Network"  
sudo nmcli delete "UUID"
```